SOFTWARE ENGINEERING LAB – 3

Architectural Style Justification

Problem Statement 001 – Smart Lab Equipment Slot Reservation Portal | Team 001 | UE24CS341A

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Architecture Selection

We chose **Layered Architecture** for the Smart Lab Equipment Slot Reservation Portal.

Architectural Choice

Layered Architecture (Presentation → Business → Data) was selected after comparing it with Microservices and Client-Server. Microservices would split reservation data across services and add network and operational overhead, and plain Client-Server would mix business rules with the user interface. Layered Architecture keeps the interface, the booking rules and the data store separate, which suits a single-lab system with moderate load.

Reason 1 – Strict consistency for concurrent bookings (FR-001, FR-003, NFR-001, NFR-002)

Every reservation passes through one business layer to a single central database with row-level locking (IReservationData). When two students request overlapping slots, only one lock succeeds, so double-booking and race conditions are prevented. Confirmed reservations and tokens are stored in one persistent store, so they survive logout, refresh and restart. Microservices would need distributed transactions to give the same guarantee.

Reason 2 – Clear separation that maps to the requirements

Each requirement has one home: FR-001 and FR-003 in the Reservation Manager, FR-002 in the Calibration Service, FR-005 and search/filter in Equipment Management, and FR-004 in the Return & Late-Return Manager. A change to calibration rules touches only the Calibration Service. The well-known drawbacks of layering, such as extra overhead and coupling between layers, are small at this scale, while the independent scaling of Microservices would add complexity without real benefit for one lab.

Security Advantage

Only the business layer can reach the database, so the browser never touches stored data directly. The external Authentication Service issues a token and role when a user logs in through IAuth, and the business layer validates that token and role on the server for every request. As a result, only authenticated students can reserve a slot (FR-001) and only authorized lab technicians can change equipment availability or calibration status (FR-005).

Performance Benefit

A reservation follows a short, direct path: Web Portal UI → Reservation Manager (with one internal call to the Calibration Service) → one database transaction. There are no extra network hops between many separate services, which helps meet the 200 ms target in NFR-001. Row-level locks are held only for the booked slot, so bookings for different equipment or time slots proceed in parallel.